

SMART HOME WATER TANK MONITORING AND AUTOMATIC CONTROL SYSTEM WITH STATISTICAL ANALYSIS

As per the 4th Semester mini project

**Bachelor of Technology
Department of Electrical Engineering**

Rupak Chatterjee

Roll no.120016244041

Aranyak Das

Roll no.120016244010

Pinaki Das

Roll no.120016244034

Pratistha Bhadra

Roll no.120016244035

Pritam Mondal

Roll no.120016244036

Under the Supervision of
Kingsuk Majumdar, Ph.D.(Engg)
Assistant Professor
Dept. of Electrical Engineering
BCREC, Durgapur



DR. B. C. ROY ENGINEERING COLLEGE

(An Autonomous Institute, Affiliated to MAKAUT)

Department of Electrical Engineering
Durgapur – 713206, West Bengal, India

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DR. B. C. ROY ENGINEERING COLLEGE

(An Autonomous Institute, Affiliated To MAKAUT)
Department of Electrical Engineering Durgapur – 713206,
West Bengal, India

CERTIFICATE OF APPROVAL

This report is hereby approved as a creditable work for 4th Sem mini project (Embedded System) on **“SMART HOME WATER TANK MONITORING AND AUTOMATIC CONTROL SYSTEM WITH STATISTICAL ANALYSIS”** carried out and presented by :

RUPAK CHATTERJEE [120016244041]

ARANYAK DAS [120016244010]

PINAKI DAS [120016244034]

PRATISTHA BHADRA [120016244035]

PRITAM MONDAL [120016244036]

in the Department of Electrical Engineering from Dr. B.C. Roy Engineering College, Durgapur under the supervision of Kingsuk Majumdar, Ph.D (Engg) as per the requirement of academic curriculum during the academic year 2025-2026.

(Signature of Supervisor)

Kingsuk Majumdar, Ph.D(Engg)

Assistant Professor

Department of Electrical Engineering

Dr. B. C. Roy Engineering College, Durgapur

(Signature of Head of Department)

Prof. (Dr.) Shibendu Mahata

Associate Professor

Head of the Department(HoD)

Department of Electrical Engineering

Dr. B. C. Roy Engineering College, Durgapur

Place: _____

Date: _____

Place: _____

Date: _____

PREFACE

This report presents the design and implementation of the project entitled **SMART HOME WATER TANK MONITORING AND AUTOMATIC CONTROL SYSTEM WITH STATISTICAL ANALYSIS**.

The project was carried out as a Fourth Semester Mini Project in the Department of Electrical Engineering at Dr. B. C. Roy Engineering College, Durgapur.

The developed prototype integrates Arduino UNO, HC-SR04 ultrasonic sensing, relay-based automatic pump control, LCD display, LED indicators, buzzer alerts, AUTO/-MANUAL switching through a toggle switch, CSV data logging, and a Python-based industrial dashboard for real-time monitoring and statistical analysis.

This report documents the complete hardware implementation, software development, testing procedure, statistical analysis, and the final outcomes achieved during the project.

ABSTRACT

Water scarcity and inefficient water management pose significant challenges in modern urban households, leading to resource wastage, increased electricity consumption, and damage to water pumping infrastructure. This project presents the design and development of a Smart Home Water Tank Monitoring and Automatic Control System that integrates embedded systems, sensor technology, and data analytics to provide an intelligent, low-cost solution for automated water level regulation.

The system employs an Arduino Uno microcontroller interfaced with an HC-SR04 ultrasonic sensor to continuously measure the water level in an overhead tank in real-time. Based on predefined threshold levels (ranging from empty to overflow prevention), the system automatically controls a water pump via a 1-channel relay module, ensuring the pump operates only when necessary. A 16×2 LCD display provides real-time visual feedback on water level status and system conditions, while LED indicators and a buzzer generate audio-visual alerts for critical states such as empty tank, filling, full tank, and overflow prevention. A manual override feature is also incorporated for user intervention when required.

Beyond real-time automation, the system incorporates data logging and statistical analysis capabilities. Water level readings, pump ON/OFF durations, refill frequencies, and operational timestamps are recorded in CSV format. This logged data is subsequently analyzed using Python (with libraries such as Pandas, Matplotlib, and PySerial) and Microsoft Excel to generate meaningful insights, including daily/weekly consumption patterns, peak usage hours, pump efficiency metrics, and potential water wastage trends. These analytics empower users to make informed decisions regarding their water consumption habits.

The system successfully demonstrates automatic pump control, overflow prevention, dry-run protection, and real-time status indication while maintaining cost-effectiveness through readily available components. Experimental validation confirms the system's reliability in preventing water wastage, reducing manual intervention, and optimizing energy usage. The project serves as a scalable foundation for smart water management and can be further enhanced with IoT integration, remote monitoring, and multi-tank support.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to everyone who supported and guided us throughout the successful completion of this Mini Project, “**Smart Home Water Tank Monitoring and Automatic Control System with Statistical Analysis.**”

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We are grateful to all the faculty members and laboratory staff of the Department of Electrical Engineering for their cooperation and for providing access to the tools, components, and laboratory infrastructure that made this project possible.

We would also like to acknowledge the collective effort, dedication, and teamwork of our project group members — **Rupak Chatterjee, Aranyak Das, Pinaki Das, Pratistha Bhadra, and Pritam Mondal** — whose combined contributions in hardware design, software development, testing, and documentation brought this project to fruition.

Finally, we express our gratitude to our families and friends for their unwavering support, patience, and motivation throughout the duration of this project.

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1. INTRODUCTION

Urban and residential water distribution systems in modern smart homes demand precise automation and analytical oversight to prevent resource depletion and physical infrastructure degradation. Traditional water tank systems suffer from continuous human oversight errors, leading to severe tank overflows, dry-running of motor pumps, and erratic power consumption patterns. By integrating embedded microcontroller architecture (Arduino Uno) with real-time distance telemetry and statistical data modeling, automated water level regulation systems provide a reliable, low-cost automation platform.

Thus, the project integrates:

- Embedded programming
- Sensor interfacing
- Electrical control
- Motor/pump control
- Instrumentation
- Data logging
- Statistical analysis
- Basic home automation

2. REQUIREMENTS

The project requires both hardware and software resources.

2.1 Hardware Requirements

- Arduino Uno
- HC-SR04 ultrasonic sensor
- 16×2 LCD display
- 1-channel relay module
- Buzzer
- LEDs
- Resistors
- Wires and connectors

- Breadboard
- Project enclosure/case
- Power supply and testing equipment

2.2 Software Requirements

- Arduino IDE 2.x
- Python 3.13
- Visual Studio Code
- Git and GitHub
- Pandas
- Matplotlib
- PySerial
- ttkbootstrap
- Pillow
- Microsoft Excel

3. OBJECTIVES

The major objectives of the project are:

Objective 1 — Automatic Water-Level Monitoring

To continuously monitor the water level of a household overhead tank using an ultrasonic sensor.

Objective 2 — Automatic Pump Control

To automatically switch the pump ON or OFF according to predefined water-level conditions.

Objective 3 — Overflow Prevention

To prevent water overflow by switching OFF the pump when the tank approaches its full level.

Objective 4 — Dry-Run Protection

To incorporate protection against unnecessary pump operation under low-water or unsuitable operating conditions.

Objective 5 — Real-Time Display

To display water-level information and system status on an LCD.

Objective 6 — Alert Generation

To provide audio-visual alerts for important tank conditions using LEDs and a buzzer.

Objective 7 — Data Logging

To record water-level information, pump operating hours, refill frequency and other operational information in CSV format.

Objective 8 — Statistical Analysis

To analyze the collected data using Python and Excel to determine average consumption, peak usage periods, pump operation and water-wastage patterns.

These objectives are aligned with the design intent outlined in Section 1.

4. WORKING PRINCIPLE AND SYSTEM OPERATION LOGIC

The system operates based on a set of predefined threshold levels. The ultrasonic sensor continuously measures the distance from the sensor to the water surface, which is then converted into a water level percentage.

Sensor Data Conversion:

- The distance is measured in centimeters using the time of flight of the ultrasonic pulse.
- Water level percentage is calculated based on the maximum and minimum possible distances (tank depth).

System Operation Logic:

The Arduino uses the following logic to control the pump and alerts:

Water Level (%)	Condition	Pump Status	LED Indication	Buzzer Status	LCD Message
0–10%	Tank Empty	ON	Red LED	Continuous	“EMPTY! PUMP ON”
10–30%	Tank Low	ON	Yellow LED	Intermittent Beep	“LOW! FILLING”
30–80%	Tank Normal	ON (if below 80%) / OFF (if above 80%)	Green LED (if ON)	No Sound	“NORMAL” / “PUMP ON”
80–95%	Tank Full	OFF	Blue LED	Success Beep (once)	“FULL! PUMP OFF”
> 95%	Overflow Prevention	OFF (Emergency)	Red LED	Alarm	“OVERFLOW! STOP”

Table 1: Water Level Monitoring and Pump Control Conditions

5. ALGORITHM

5.1 Flowchart for the System Operation

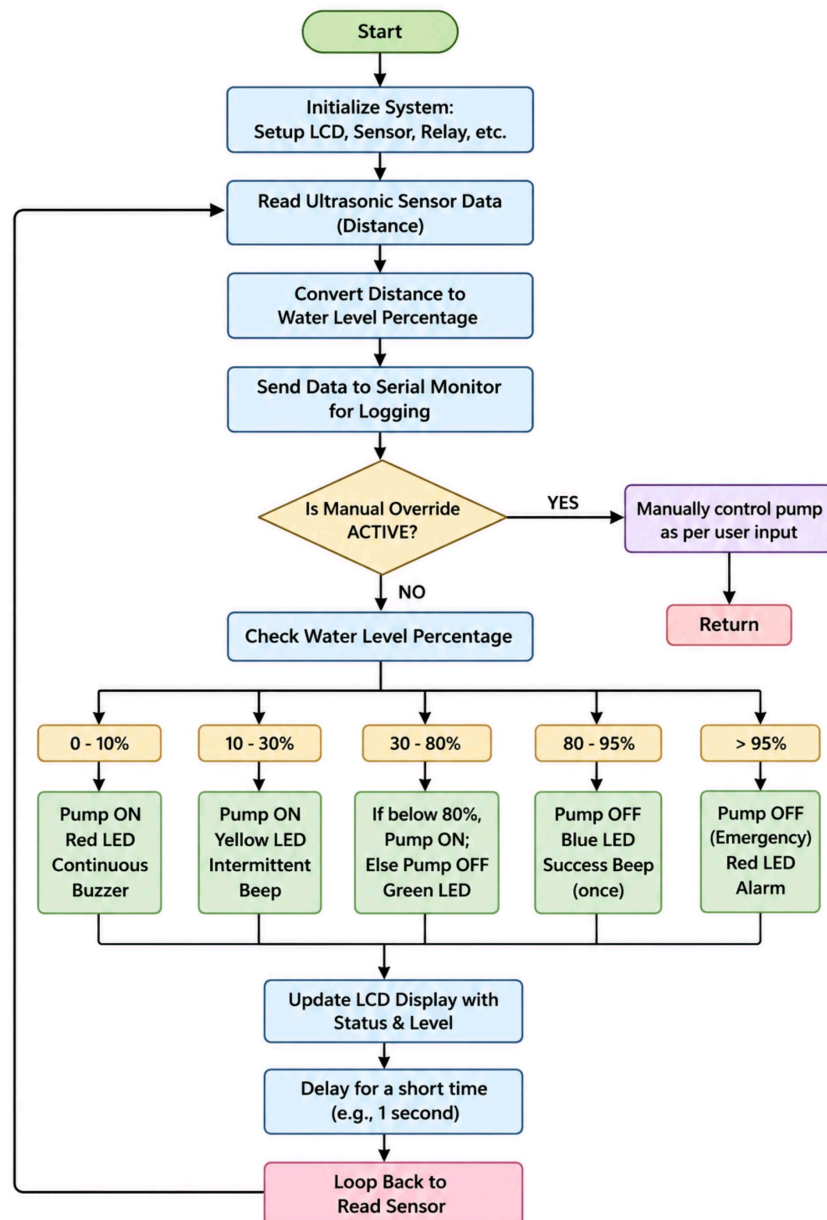


Figure 1: Ultrasonic Water Level Control Flowchart
(Taken from google)

ALGORITHM FOR SYSTEM OPERATION

Listing 1: Algorithm for Water Level Sensing and Pump Control

```
1  ALGORITHM: Main Water Level Sensing and Pump Control Loop
2
3  START
4
5  STEP 1: Initialize system
6      - Setup LCD display
7      - Setup ultrasonic sensor pins
8      - Setup relay pin
9      - Setup LED and buzzer pins
10     - Setup manual override switch pin
11
12  LOOP forever:
13
14     STEP 2: Read ultrasonic sensor data (measure distance to water surface)
15
16     STEP 3: Convert measured distance to Water Level Percentage
17
18     STEP 4: Send level and status data to Serial Monitor for logging
19
20     STEP 5: Check if Manual Override is ACTIVE
21         IF Manual Override == YES THEN
22             Manually control pump as per user input
23             RETURN (exit this cycle, skip automatic logic)
24         ELSE
25             Continue to STEP 6
26
27     STEP 6: Check Water Level Percentage and apply control logic:
28
29         IF level% is between 0% and 10% THEN
30             Pump = ON, LED = Red, Buzzer = Continuous
31
32         ELSE IF level% is between 10% and 30% THEN
33             Pump = ON, LED = Yellow, Buzzer = Intermittent Beep
34
35         ELSE IF level% is between 30% and 80% THEN
36             IF level% < 80% THEN Pump = ON
37             ELSE Pump = OFF
38             LED = Green
39
40         ELSE IF level% is between 80% and 95% THEN
41             Pump = OFF, LED = Blue, Buzzer = Success Beep (single)
42
```

```
43         ELSE IF level% > 95% THEN
44             Pump = OFF (Emergency Stop), LED = Red, Buzzer = Alarm
45
46     STEP 7: Update LCD Display with current status and level%
47
48     STEP 8: Delay for a short time (e.g., 1 second)
49
50     STEP 9: Go back to STEP 2 (repeat loop)
51
52 END LOOP
```

6. PROJECT REPOSITORY

The complete source code, documentation, circuit diagrams, Python dashboard, GitHub Actions workflows, and project assets are maintained in the project's GitHub repository.

Repository:



Figure 2: GitHub Repository QR Code

The repository includes:

- Arduino firmware
- Python industrial dashboard
- Circuit diagrams
- Hardware photographs
- Statistical analysis documentation
- GitHub Actions validation workflows
- Project documentation

6.1 Flowchart for Statistical Data Analysis

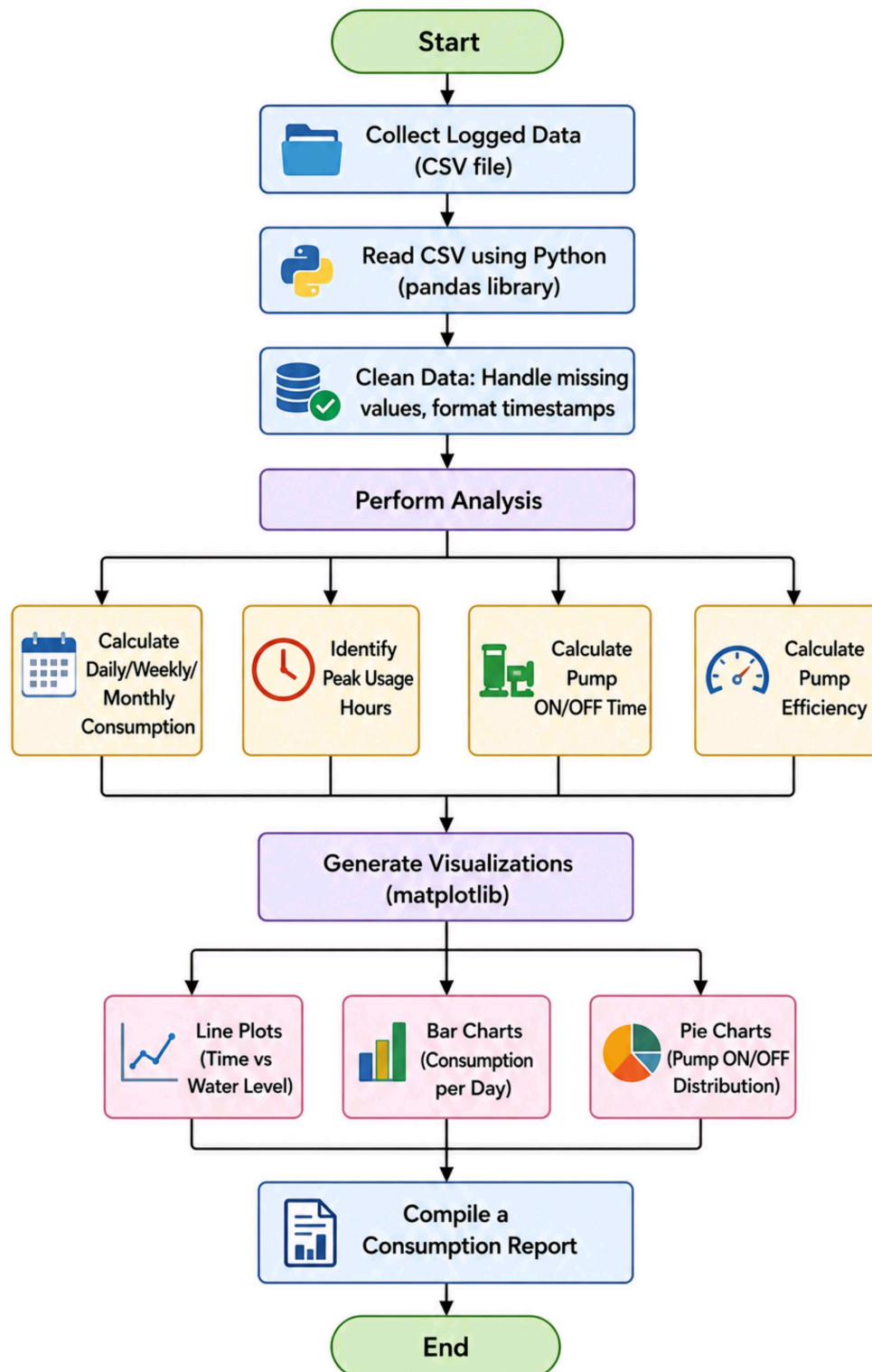


Figure 3: Data Analysis and Reporting Flowchart
(Taken from google)

Algorithm for Statistical Data Analysis

Listing 2: Algorithm for Statistical Data Analysis

```
1  ALGORITHM: Water Consumption Data Analysis and Reporting
2
3  START
4
5  STEP 1: Collect logged data (tank_log.csv file) from the monitoring system
6
7  STEP 2: Read the CSV file into a pandas Data Frame
8
9  STEP 3: Clean the data
10         - Handle missing or corrupted values
11         - Format and parse timestamp column into proper date/time objects
12
13  STEP 4: Perform analysis on the cleaned data
14         4.1: Calculate Daily / Weekly / Monthly water consumption
15         4.2: Identify Peak Usage Hours (hours with highest pump activity)
16         4.3: Calculate total Pump ON time and Pump OFF time
17         4.4: Calculate Pump Efficiency (useful run time vs total run time)
18
19  STEP 5: Generate visualizations using matplotlib
20         5.1: Line Plot -> Water Level (%) vs Time
21         5.2: Bar Chart -> Water Consumption per Day
22         5.3: Pie Chart -> Pump ON/OFF Time Distribution
23
24  STEP 6: Compile all results and charts into a Consumption Report
25         (export as text/HTML/Excel summary)
26
27  END
```

7. LIBRARIES FOR THE PROGRAM TO RUN

- **pyserial** – Reads live Arduino data through the serial port.
- **pandas** – Performs CSV parsing and statistical calculations.
- **matplotlib** – Generates live trend graphs.
- **ttkbootstrap** – Creates the industrial dark dashboard UI.
- **Pillow** – Handles dashboard icons and images.
- **tkinter** – Builds the desktop GUI.
- **csv** – Stores sensor readings in CSV format.

- **datetime** – Generates timestamps for every reading.

8. ADVANTAGES AND LIMITATIONS

8.1 Advantages

- **Water Conservation:** Prevents overflow and wastage, promoting efficient water usage.
- **Energy Efficiency:** Automatically turns off the pump when full, saving electricity.
- **Convenience:** Eliminates the need for manual monitoring and pump operation.
- **Protection:** Prevents dry-running, which can damage the pump motor.
- **Data-Driven Insights:** Provides statistical analysis to help users understand and modify their consumption habits.
- **Cost-Effective:** Uses readily available, affordable components.

8.2 Limitations

- **Tank Depth:** Ultrasonic sensors can be affected by tank geometry and reflections in deep or irregularly shaped tanks.
- **Sensitivity:** Sensor readings can be prone to errors due to dust, humidity, or temperature variations.
- **Single Source:** The system controls only one tank and one pump; scaling it for multi-tank or complex buildings would be more challenging.
- **Power Dependency:** The system relies on a stable power supply.

9. IMPLEMENTATION AND RESULT

The project was successfully implemented using Arduino UNO and tested under different water level conditions. The ultrasonic sensor measured the distance from the sensor to the water surface, while the Arduino calculated the corresponding water level percentage.

The LCD displayed real-time water level and pump status. The relay controlled the pump automatically according to preset thresholds. Visual indication was provided through LEDs, while the buzzer generated an alert during low-water conditions.

The Python dashboard received serial data continuously, stored readings in CSV format, and generated live graphs for statistical analysis.

9.1 CIRCUIT DIAGRAM

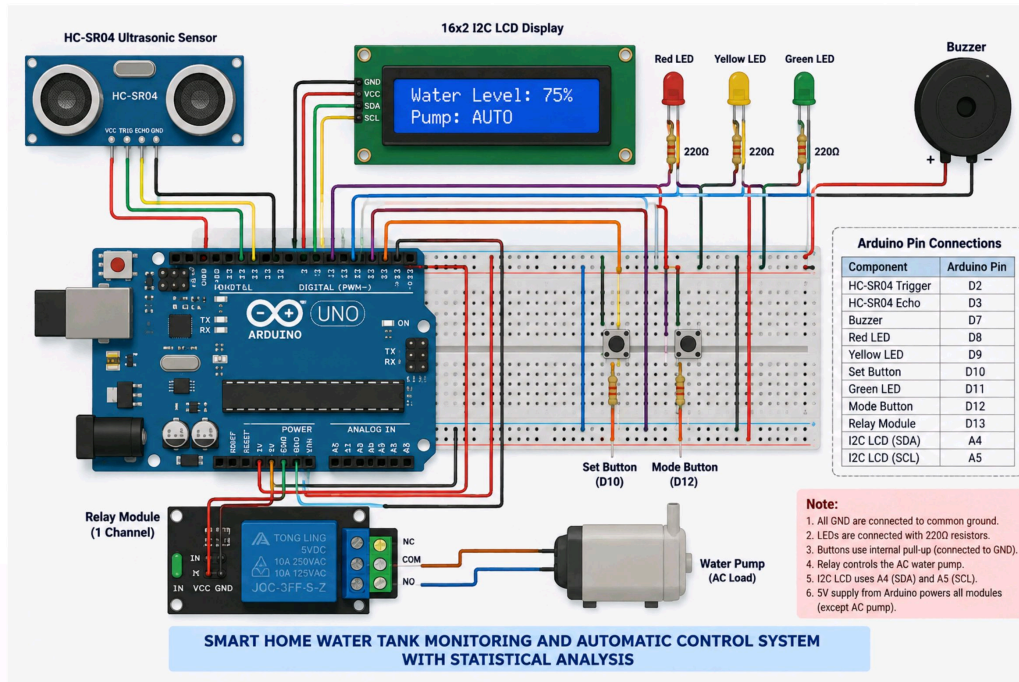


Figure 4: Circuit diagram for the Smart Water Tank
(Taken from google)

9.2 WATER LEVEL MONITORING

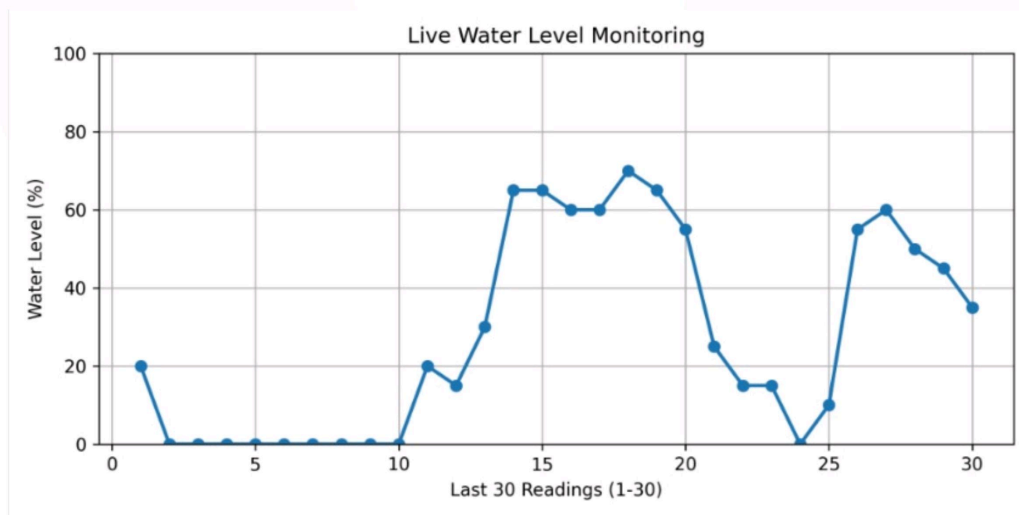


Figure 5: Water level Monitoring

9.3 Python Dashboard

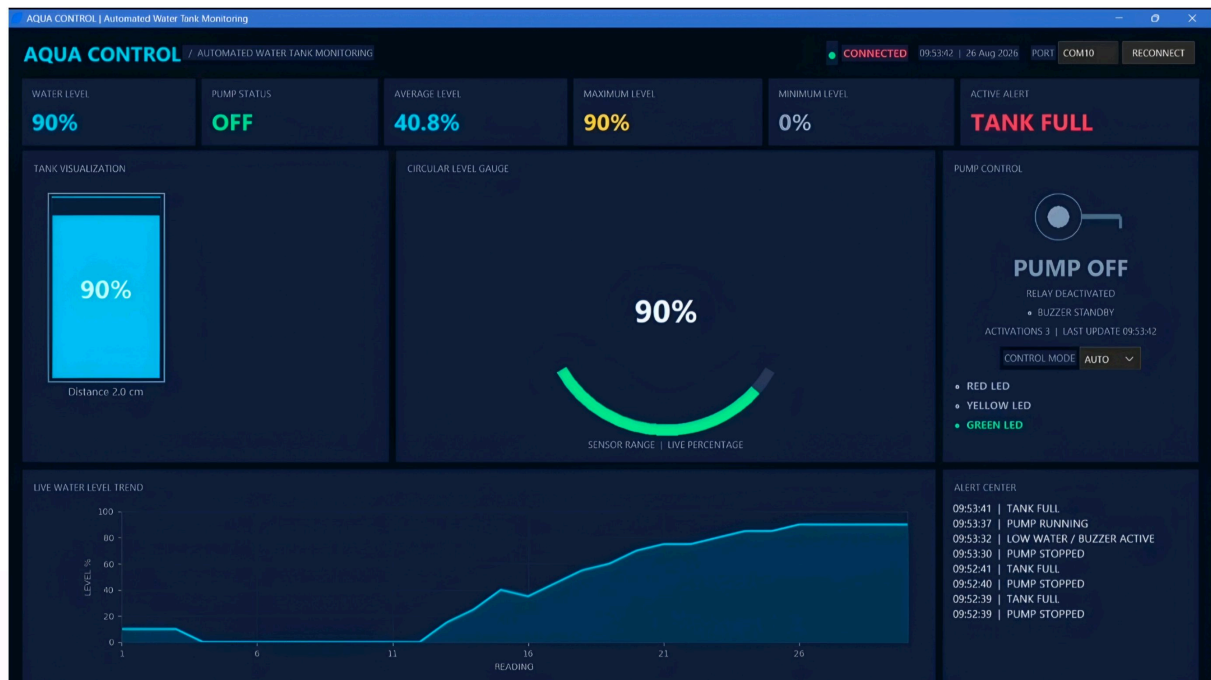


Figure 6: Industrial Python Dashboard with Live Monitoring

Dashboard Output



Figure 7: Scan to view the Dashboard & Workflow Demonstration

9.4 Statistical Analysis Results

The logged CSV data was analyzed using Python. The dashboard continuously calculated the average water level, maximum level, minimum level, pump activation count and live trend analysis.

- Average Water Level
- Maximum Water Level
- Minimum Water Level
- Pump Activation Count

- Live Trend Analysis

Average Level	82.5%
Maximum Level	95%
Minimum Level	30%
Pump Activations	3
Current Mode	AUTO

Table 2: Sample Statistical Output

9.5 HARDWARE IMPLEMENTATION

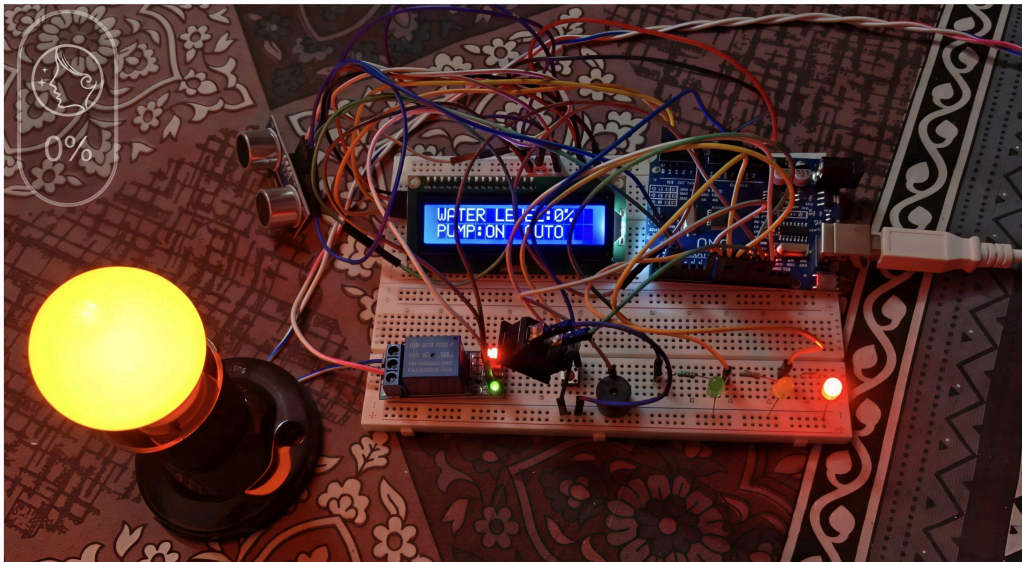


Figure 8: Hardware Implementation of the Smart Water Tank Control System

10. SYSTEM TESTING

The prototype was tested under different operating conditions.

Test Case	Result
Tank Empty	Pump turned ON automatically.
Tank Full	Pump turned OFF automatically.
Manual Mode	Pump toggled using push button.
AUTO Switch	Mode changed successfully.
LED Indicators	Correctly indicated water level.
Buzzer Alert	Activated at low water level.
CSV Logging	Live data stored successfully.
Dashboard	Real-time updates received.

Table 3: System Validation

11. CONCLUSION

The Smart Home Water Tank Monitoring and Automatic Control System provides an efficient and reliable approach to monitoring and managing household water levels. The system integrates an Arduino Uno, ultrasonic sensor, LCD display, relay module, LEDs, buzzer, and water pump to continuously monitor the tank level and automatically control the pumping operation according to predefined water-level conditions.

The automatic control mechanism helps prevent water overflow and reduces the need for continuous manual monitoring. The use of visual and audio indicators provides clear information about important tank conditions, while the LCD displays the current system status to the user. The system also incorporates protection against unsuitable pump operation, thereby improving the overall reliability of the setup.

In addition to real-time monitoring and automatic control, the project includes data logging and statistical analysis using Python and Excel. The collected information can be used to study water consumption, pump operating patterns, refill frequency, and potential water-wastage trends. This combination of automation and data analysis makes the system useful not only for controlling the water tank but also for understanding and improving water usage.

Overall, the project demonstrates how embedded systems, sensor interfacing, electrical control, and statistical analysis can be combined to develop a practical smart-home application. The proposed system provides a low-cost foundation for automated water management and can be further enhanced in the future by incorporating remote monitoring, wireless connectivity, multiple-tank support, and more advanced data analysis techniques.

12. FUTURE SCOPE

While the current Smart Home Water Tank Monitoring and Automatic Control System successfully demonstrates automated water level regulation, pump control, and statistical analysis, there are numerous avenues for enhancement and expansion. These improvements can elevate the system from a standalone solution to a fully integrated, intelligent, and scalable water management ecosystem.

- IoT integration with ESP8266/ESP32 for remote monitoring and control via smart-phone
- Push notifications and SMS alerts for critical tank conditions
- Mobile application development for real-time data visualization and system control
- Multi-tank support for larger residential and commercial buildings
- Integration of flow sensors for volume-based consumption tracking
- Integration of pressure sensors for leak detection
- Integration of water quality sensors (pH, TDS, turbidity)
- Machine learning for consumption prediction and anomaly detection
- Predictive maintenance alerts for pump failures
- Solar power integration for energy independence
- Cloud data storage for long-term trend analysis
- Voice control integration with Amazon Alexa / Google Home
- Automated billing system for apartment complexes
- Compact PCB design for commercialization
- Integration with smart irrigation systems

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